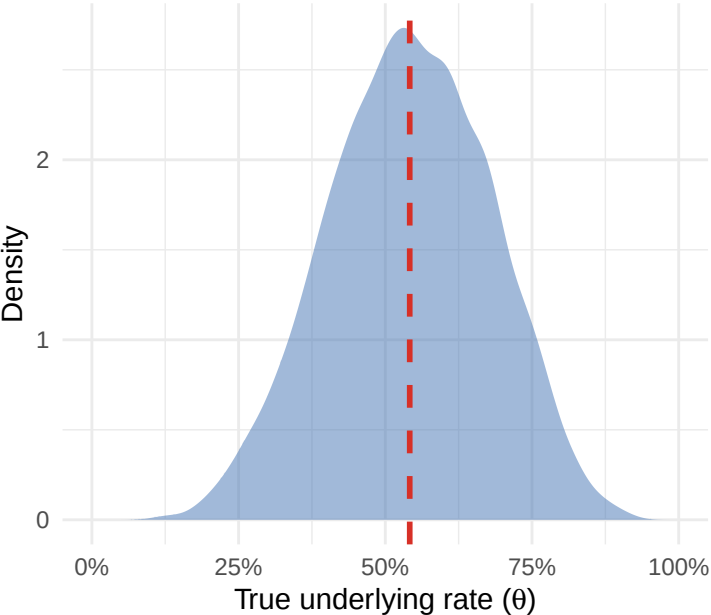


How Uncertainty Propagates: Beta-Binomial vs Binomial

When an agent makes multiple choices, uncertainty about the true rate creates a wider distribution of outcomes.

Step 1: Uncertainty in Belief (.)

Beta(6.5, 5.5) | Dashed line is $E[.] = 54\%$



Step 2: Choices out of 20 Trials

Notice the wider spread (fatter tails) of the Beta-Binomial

■ Beta-Binomial (Uncertain .) ■ Standard Binomial (Fixed .)

